p: 11 Simulate Static Routing Configuration using CISCO Packet Tracer

Static routing are the routes you manually add to the router's routing Table. The process of adding static routes to the routing table is known as Static routing

Setting up a practice lab

Create a packet tracer lab as shown in the following image or download the following Pre-created lab and load it on Packet Tracer

Packet Tracer lab with Initial IP configu

#ip route 30.0.0.0 255.00.0 20.0.0.2

#ip route 30.0.0.0 255.0.0.0 40.0.0.2

Creating, adding, verifying Static route

Routes automatically learn their connect networks We only need to add route for the network that are not available d

the route's interface.

| Route | Available network on local interface | Networks available on the router |
|---|---|---|
| Router0 | 10.0.0.0/8<br>20.0.0.0/8<br>40.0.0.0/8 | 30.0.0.0/8<br>50.0.0.0/8 |
| Router1 | 20.0.0.0/8<br>30.0.0.0/8 | 10.0.0.0/8<br>40.0.0.0/8 |
| Router 2 | 40.0.0.0/8<br>50.0.0.0/8 | 10.0.0.0/8<br>20.0.0.0/8<br>30.0.0.0/8 |

Router 0 requirements

- create two route for network 30.0.0.0/8 and configure the first route (via Router) as the main rout for host 30.0.0.100/8 and configure the first route (via Router 2) as the main route and second route as backup
- create two route for network 50.0.0.0/8 and configure the first route as the main route and second route via Router as a back up route.

Verifying Static routing

on Router(1), we configured two routes for network 30.0.0.0/8. The route are via Router as the main route and the second route as the backup route. We can verify this configure in two ways

By sending ping request to PC of network 30.0.0.0/8 and tracing the packet they take the reach the network 30.0.0/8. Lat y at you can tracert comma on PC of network 10.0.0.0/8

The following link provide the configured Packet tracker lab of the above

Deleting a Static route

To delete a Static route use the following

- Use the Show ip route Static command to print all Static routes
- Note down the route you want to delete
- Use the no ip route command to delete the route

Laptop1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>

C:\>ping 192.168.10.101

Pinging 192.168.10.101 with 32 bytes of data:

Reply from 192.168.10.101: bytes=32 time=61ms TTL=128
Reply from 192.168.10.101: bytes=32 time=42ms TTL=128
Reply from 192.168.10.101: bytes=32 time=51ms TTL=128
Reply from 192.168.10.101: bytes=32 time=49ms TTL=128

Ping statistics for 192.168.10.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 42ms, Maximum = 61ms, Average = 50ms

C:\>
```

## Experiment 11:

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#int g0/0
Router(config-if)#ip address 192.169.1.1 255.255.2550
                                         ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.169.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#int g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

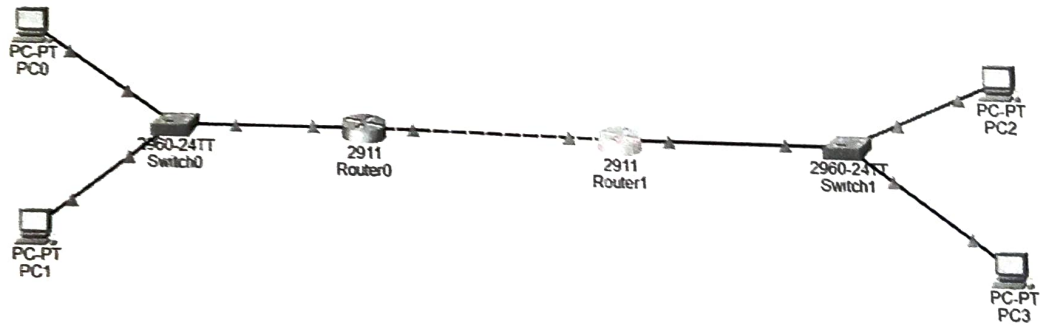
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.2

Router#
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy run-conf startup-conf
                 ^
% Invalid input detected at '^' marker.

Router#copy run-config startup-config
                 ^
% Invalid input detected at '^' marker.

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.0.0.1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0

Router#
```

## RIP_Protocol:

Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#route rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#interface g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.253
Bad mask 0xFFFFFFFD for address 10.0.0.2
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#router rip
              ^
% Invalid input detected at '^' marker.

Router#version 2
              ^
```

```
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#ver 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#copy running-config startup-config

% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#
```

Copy Paste

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

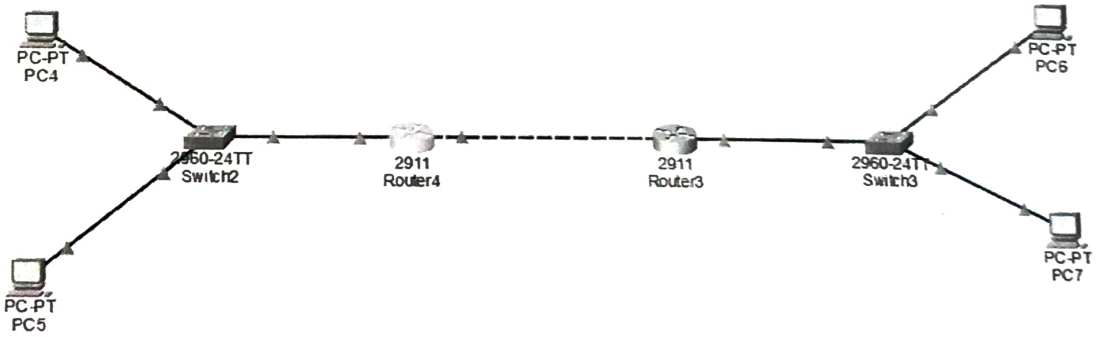
Gateway of last resort is not set

     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0

Router#
```

Copy Paste

RIP_Dynamic_Routing_Protocol



Initial IP configuration

| Device | Interface | IP configuration | connected in th |
|---|---|---|---|
| PC 0 | Fast Ethernet | 10.0.0.2/8 | Router 0 |
| Router 0 | S 0/0/01 | 192.168.1.254/30 | Router 2's S0/01 |
| Router 1 | S0/0/0 | 192.168.1.256/30 | Router 1's S0/0/0 |
| Router 1 | S0/0/1 | 192.168.1.24/130 | Router 2 S0/0/1 |
| Router 2 | S0/0/1 | 192.168.1.25/130 | Router 0's S0/0/1 |
| PC 1 | Fast Ethernet | 20.0.0.2/30 | Router 2' Fa 0/1 |

Assign IP addersstoPC

Double click PCs and click Desktop menu ite and click IP configuraturation. Assign IPadd referring the above table

Assign IP address to interface of router

Double click Router() and click CLI and Press Enter key to access the Commandprom of routers()

Result: Thus, Static routes were Successfully configured and verified.